

Companion to Bird & de Moor, Algebra of Programming

§3. Applications

§3.1. Banana-split

§3.1.1. Laws from Relation Algebra

§3.1.1.1. Adjunctions

(3.1.1.1a)

$F \dashv G$	F 's type	monad FG	$1 \in FG$	$GF \in 1$	F preserves $\cup$	G preserves $\cap$	$FGF=F$	$GFG=G$
$\triangleleft \dashv \triangleleft$	$A \rightarrow A \otimes A$	$\triangleleft \triangleright = 1$	$1 \in \triangleleft \triangleright$	$\triangleright \triangleleft \in 1$	$(R \cup S) \triangleleft = R \triangleleft \cup S \triangleleft$	$(R \cap S) \triangleright = R \triangleright \cap S \triangleright$	$\triangleleft \triangleright \triangleleft = \triangleleft$	$\triangleright \triangleleft \triangleright = \triangleright$
$\multimap \dashv \multimap$	$A \rightarrow \mathbb{I}$	$\multimap \multimap = T$	$1 \in \multimap \multimap$	$\multimap \multimap \in 1$	$(R \cup S) \multimap = R \multimap \cup S \multimap$	$(R \cap S) \multimap = R \multimap \cap S \multimap$	$\multimap \multimap \multimap = \multimap$	$\multimap \multimap \multimap = \multimap$
$\circ \dashv \circ$	$(A \rightarrow B) \rightarrow (B \rightarrow A)$	$R^\circ \circ = R$	$R^\circ \circ = R$	$R^\circ \circ = R$	$(R \cup S)^\circ = S^\circ \cup R^\circ$	$(R \cap S)^\circ = R^\circ \cap S^\circ$	$R^\circ \circ = R$	$R^\circ \circ = R$
$\multimap \triangleleft \dashv \triangleright \multimap$	$(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$	1	$(\multimap \triangleleft \otimes 1) = 1$ $(1 \otimes \triangleright \multimap) = 1$	$(1 \otimes \multimap \triangleleft) = 1$ $(\triangleright \multimap \otimes 1) = 1$	$=$	$=$	$=$	$=$
$\triangleright \multimap \dashv \multimap \triangleleft$	$(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$	1	$(1 \otimes \multimap \triangleleft) = 1$ $(\triangleright \multimap \otimes 1) = 1$	$(\multimap \triangleleft \otimes 1) = 1$ $(1 \otimes \triangleright \multimap) = 1$	$=$	$=$	$=$	$=$
$\Delta \dashv \cap$	$(A \rightarrow B) \rightarrow (A \rightarrow B)^2$	$R \cap R = R$	$R \cap R = R$	$R \cap S \in R$	$\Delta(R \cup S) = \Delta R \cup \Delta S$	$(R \cap T) \cap (S \cap U) = (R \cap S) \cap (T \cap U)$	$R \cap R = R$	$R \cap R = R$
$\cup \dashv \Delta$	$(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$	$(R, S) \mapsto (R \cup S, R \cup S)$	$R \in R \cup S$	$R \cup R = R$	$(R \cup T) \cup (S \cup U) = (R \cup S) \cup (T \cup U)$	$\Delta(R \cap S) = \Delta R \cap \Delta S$	$R \cup R = R$	$R \cup R = R$
$\emptyset \dashv !$	$\{*\} \rightarrow (A \rightarrow B)$			$\emptyset \in R$				
$\cdot S \dashv / S$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	S/S	$R \in (RS)/S$	$(T/S) S \in T$	$(S \cup T)R = SR \cup TR$	$(R \cap R')/S = (R/S) \cap (R'/S)$	$((RS)/S)S = RS$	$((T/S)S)/S = T/S$
$S \cdot \dashv S \setminus$	$(B \rightarrow C) \rightarrow (A \rightarrow C)$	$S \setminus S$	$R \in S \setminus (SR)$	$S(S \setminus T) \in T$	$R(S \cup T) = RS \cup RT$	$S \setminus (R \cap R') = (S \setminus R) \cap (S \setminus R')$	$S(S \setminus (SR)) = SR$	$S \setminus (S(S \setminus T)) = S \setminus T$
$R \cap \dashv R \Rightarrow$	$(A \rightarrow B) \rightarrow (A \rightarrow B)$	$X \mapsto R \Rightarrow (X \cap R)$	$X \in R \Rightarrow (X \cap R)$	$R \cap (R \Rightarrow Y) \in Y$	$R \cap (X \cup Y) = (R \cap X) \cup (R \cap Y)$	$R \Rightarrow S \cap T = (R \Rightarrow S) \cap (R \Rightarrow T)$	$R \cap (R \Rightarrow (X \cap R)) = X \cap R$	$R \Rightarrow (R \cap (R \Rightarrow Y)) = R \Rightarrow Y$
$\mathcal{D} \dashv \cdot T$	$(A \rightarrow B) \rightarrow \text{Cor } A$	$R \mapsto (\mathcal{D}R)T$	$R \in (\mathcal{D}R)T$	$\mathcal{D}(AT) \in A$	$\mathcal{D}(R \cup S) = \mathcal{D}R \cup \mathcal{D}S$	$(A \cap B)T = AT \cap BT$	$\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$	$(\mathcal{D}(AT))T = AT$
$\mathcal{R} \dashv T \cdot$	$(A \rightarrow B) \rightarrow \text{Cor } B$	$R \mapsto T(\mathcal{R}R)$	$R \in T(\mathcal{R}R)$	$\mathcal{R}(TA) \in A$	$\mathcal{R}(R \cup S) = \mathcal{R}R \cup \mathcal{R}S$	$T(A \cap B) = TA \cap TB$	$\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$	$T(\mathcal{R}(TA)) = TA$
$\cdot f \dashv \cdot f^\circ$	$(A \rightarrow B) \rightarrow (A \rightarrow C)$	$f f^\circ$	$\text{Map}(f) \Rightarrow 1 \in f f^\circ$	$f^\circ f \in 1$	$(S \cup T)R = SR \cup TR$	$\text{Simple}(g) \Rightarrow (X \cap Y)g^\circ = Xg^\circ \cap Yg^\circ$	$f f^\circ f = f$	$f^\circ f f^\circ = f^\circ$
$f^\circ \cdot \dashv f \cdot$	$(A \rightarrow C) \rightarrow (B \rightarrow C)$	$f f^\circ$	$\text{Map}(f) \Rightarrow 1 \in f f^\circ$	$f^\circ f \in 1$	$R(S \cup T) = RS \cup RT$	$f(X \cap Y) = fX \cap fY$	$f f^\circ f = f$	$f^\circ f f^\circ = f^\circ$
$i \dashv E$	$\text{Map} \dashv \text{Rel}$	E	$\frac{1}{\exists} : A \rightarrow EA$	$\exists : EB \rightarrow B$	—	—	$\frac{\exists}{\exists} = 1$	$\frac{R}{\exists} \exists = R$
$\cdot \exists \dashv \frac{\exists}{\exists}$	$\text{Map}(A, EB) \rightarrow (A \rightarrow B)$	1	$\frac{f \exists}{\exists} = f$	$\frac{R \exists}{\exists} = R$	—	—	$\frac{f \exists}{\exists} = f \exists$	$\frac{R \exists}{\exists} = R \exists$
$\frac{\exists}{\exists} \dashv \cdot \exists$	$(A \rightarrow B) \rightarrow \text{Map}(A, EB)$	1	$\frac{R \exists}{\exists} = R$	$\frac{f \exists}{\exists} = f$	—	—	$\frac{R \exists}{\exists} = R \exists$	$\frac{f \exists}{\exists} = f \exists$

§3.1.1.2. Composing adjunctions

(3.1.1.2a)

	$\cdot T$	$T \cdot$	$\cdot g$	$g \cdot$	Tn	$\circ$	$\circ \triangleleft$	Δ
$\cdot S$	$R/S_1 \ S_2 = R/S_2/S_1$	$S \setminus (R/W) = (S \setminus R)/W$	$R/(Sg) = (Rg^\circ)/S$	$\text{Map}(f) \Rightarrow f(R/S) = fR/S$	—	$(Y/S)^\circ = S^\circ \setminus (Y^\circ)$	$(RS)^\wedge = R^\wedge (S \otimes 1)$	$(RnR')/S = (R/S) \cap (R'/S)$
$S \cdot$	$S \setminus (R/W) = (S \setminus R)/W$	$S_1 \ S_2 \setminus R = S_2 \setminus (S_1 \setminus R)$	$S \setminus (Rg^\circ) = (S \setminus R)g^\circ$	$(g^\circ S) \setminus R = S \setminus (gR)$	—	$(S \setminus Y)^\circ = Y^\circ/S^\circ$	$((S \otimes 1)R)^\wedge = SR^\wedge$	$S \setminus (RnR') = (S \setminus R) \cap (S \setminus R')$
$\cdot f$	$\text{Map}(f) \Rightarrow R/fS = (R/S)f^\circ$	$T \setminus (Rf^\circ) = (T \setminus R)f^\circ$	$(RS)^\circ = S^\circ R^\circ$			$(RS)^\circ = S^\circ R^\circ$		$\text{Simple}(g) \Rightarrow (XnY)g^\circ = Xg^\circ n Yg^\circ$
$f^\circ \cdot$	$\text{Map}(f) \Rightarrow f(R/S) = fR/S$	$(Tf^\circ) \setminus R = f(T \setminus R)$	—	$(RS)^\circ = S^\circ R^\circ$		$(RS)^\circ = S^\circ R^\circ$		$f(T_1 n T_2) = fT_1 n fT_2$
Rn	—	—	—	—	$R \Rightarrow S \Rightarrow T = RnS \Rightarrow T$	$(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$	—	$R \Rightarrow SnT = (R \Rightarrow S) \cap (R \Rightarrow T)$
$\circ$	$(Y/T)^\circ = T^\circ \setminus (Y^\circ)$	$(T \setminus Y)^\circ = Y^\circ/T^\circ$	$(RS)^\circ = S^\circ R^\circ$	$(RS)^\circ = S^\circ R^\circ$	$(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$	$R^\circ \circ = R$	$(R^\wedge)^\circ = (R^\circ \otimes 1)$ $(1 \otimes \triangleright \dashv \circ)$ both bends: the $\circ$ section's display	$(T_1 n T_2)^\circ = T_1^\circ n T_2^\circ$
$\circ \triangleleft$	$(RT)^\wedge = R^\wedge (T \otimes 1)$	$((T \otimes 1)R)^\wedge = TR^\wedge$	—	—	—	$(R^\wedge)^\circ = (R^\circ \otimes 1)$ $(1 \otimes \triangleright \dashv \circ)$ both bends: the $\circ$ section's display	—	—
u	$(S \cup T)R = SR \cup TR$	$R(S \cup T) = RS \cup RT$	$(S \cup T)R = SR \cup TR$	$R(S \cup T) = RS \cup RT$	$Rn(S \cup T) = RnS \cup RnT$	$(R \cup S)^\circ = S^\circ \cup R^\circ$		
$\emptyset$	$\emptyset R = \emptyset$	$R\emptyset = \emptyset$	$\emptyset R = \emptyset$	$R\emptyset = \emptyset$	$Rn\emptyset = \emptyset$	$\emptyset^\circ = \emptyset$		

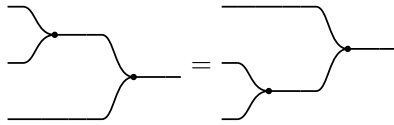
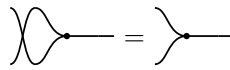
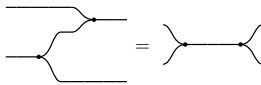
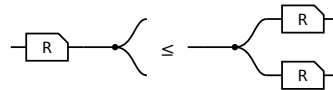
§3.1.1.3. Adjoint triples

(3.1.1.3a)

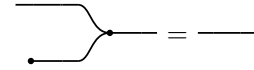
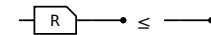
$$\begin{aligned} \cdot f \dashv \cdot f^\circ \dashv / f^\circ \\ f^\circ \cdot \dashv f \cdot \dashv f \setminus \end{aligned}$$

§3.1.1.4. Copy and merge

(3.1.1.4a)

 $\triangleright$ associative $\triangleright$ commutative

$$R \triangleleft \leq \triangleleft (R \otimes R)$$

 $\circ$ is its unit

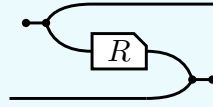
$$R \circ \leq \circ$$

Frobenius, one half — the other is its $\circ$ §3.1.1.5. $\circ$ is a contravariant 2-functor

(3.1.1.2a)

(3.1.1.5a)

$\circ$ is primitive, part of the data the first section lists. It turns both of R 's wires round, and the Frobenius structure **DRAWS** that — the picture and the formula below are R° , not its definition:



$$R^\circ = (\circ \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \multimap \circ)$$

where $\circ \triangleleft : \mathbb{I} \rightarrow a \otimes a$ opens a pair of wires out of nothing and $\triangleright \circ : b \otimes b \rightarrow \mathbb{I}$ closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: $\triangleleft$ and $\multimap$ are $\triangleright^\circ$ and $\circ^\circ$. The laws of $\circ$ are that it is a contravariant 2-functor $\circ : \mathcal{C}^{\circ p} \rightarrow \mathcal{C}$:

- (i) $1^\circ = 1$ (ii) $(RS)^\circ = S^\circ R^\circ$ (iii) $(R \otimes S)^\circ = R^\circ \otimes S^\circ$ (iv) $R \leq S$ implies $R^\circ \leq S^\circ$

§3.1.1.6. Meet and composition

(3.1.1.6a)

semi-distributivity, and what supplies it	picture
$R (S \cap T) \sqsubseteq RS \cap RT$ — the lax copy law. Equality exactly when R is single valued: the Maps section's $F(R \cap S) = FR \cap FS$.	

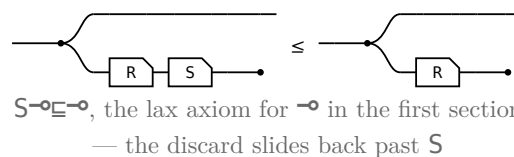
§3.1.1.7. Domain and range

(3.1.1.7a)

$\text{dom}(R) \triangleq 1 \cap R R^\circ$
$\text{Coreflexive}(X) \implies \text{dom}(R) \sqsubseteq X \iff R \sqsubseteq X R$
$\text{dom}(RS) \sqsubseteq \text{dom}(R)$
$\text{dom}(R \cap S) = 1 \cap S R^\circ$
$R \text{ entire} \iff \text{Entire}(R) \triangleq \text{dom}(R) = 1$
$R \text{ simple} \iff \text{Simple}(R) \triangleq R^\circ R \sqsubseteq 1$
$R \text{ a map} \iff \text{Map}(R) \triangleq \text{Entire}(R) \wedge \text{Simple}(R)$
$\text{Entire}(R) \wedge \text{Entire}(S) \implies \text{Entire}(RS)$ $\text{Simple}(f) \wedge \text{Simple}(g) \implies \text{Simple}(fg)$ $\text{Map}(f) \wedge \text{Map}(g) \implies \text{Map}(fg)$
$\text{Entire}(RS) \implies \text{Entire}(R)$

§3.1.1.8. Sliding the discard

(3.1.1.8a)



§3.1.1.9. Division

(3.1.1.5a)

$X \in R/S \iff XS \in R$ X is any x -to- y pairing; one that only pairs x with a y such that x admires everyone y hates lies inside R/S , and R/S is the largest such.	
$X \in S \setminus R \iff SX \in R$ The mirror — divide on the left when x comes first.	
$(R/S)S \in R$ There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at $S=\emptyset$: R/S is everyone, $(R/S)S=\emptyset$.	
$S \setminus (S \setminus R) \in R$ The mirror.	
associate: $R/(S_1 S_2) = (R/S_2)/S_1$ A friend's enemy is two hops: divide by the far end first.	
$(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$ The mirror.	
maps: $f \setminus (R/S) = (fR)/S$ Rename x before or after dividing — the licence to write fR/S .	
$R/(fS) = (R/S) f^\circ$ Rename y : a map leaves a denominator as f° outside the box.	
$(R/S)(S/W) \in R/W$ Someone who admires all of a hate-set that already covers everyone z works for admires those people too.	
$1 \in R/R$ R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.	
$(R/R)(R/R) = R/R$ R/R is the preorder admires at least as much as , and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $1 \in R/R$ above it is an equality.	
$(R/R)R = R$ Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.	
$R/1 = R$ Dividing by 1 : p 's set is just $\{p\}$, so admiring all of it is admiring p .	
$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ Admiring a combined hate-set is admiring each set in full.	
$S \setminus (R/W) = (S \setminus R)/W$ Which is why $S \setminus R/W$ needs no bracket.	

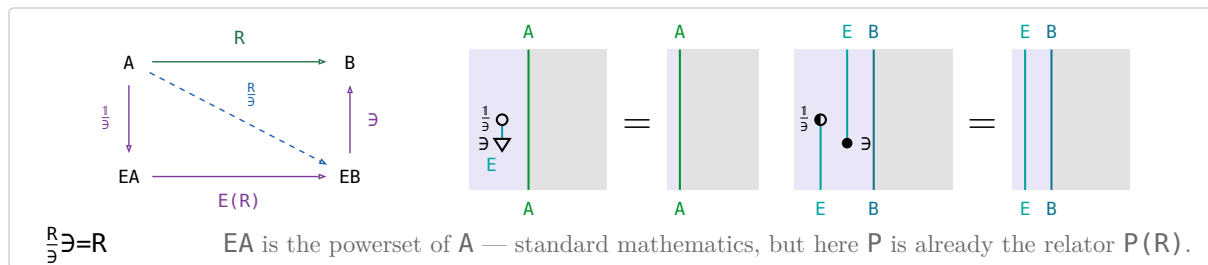
§3.1.1.10. Power allegories

(3.1.1.10a)

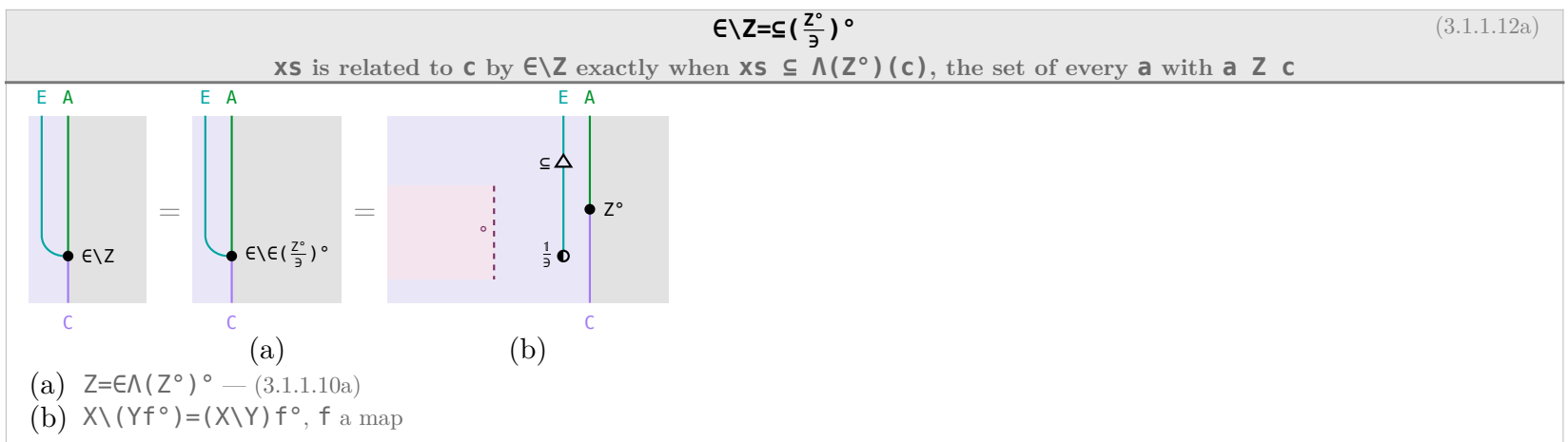
law	the reading
1 $\exists_R \square = R\square, \exists_R = \exists_R \square$	One $\exists$ per object, not per arrow.
2 $\exists$ is thick	Comprehension: every x has a set of exactly the people x admires.
3 $\frac{R}{\exists} : A \rightarrow EB$, for $R : A \rightarrow B$	convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$
4 $\text{Map}(\frac{R}{\exists})$	
5 $\frac{R}{\exists} \exists = R$	reading the chosen set back through $\exists$ returns the relation.
6 $\text{Simple}(F) \Rightarrow F \sqsubseteq \frac{F}{\exists}$	A partial choice of sets is inside the total one.
7 the singleton map is monic: $\frac{1}{\exists}$ $(\frac{1}{\exists})^\circ \sqsubseteq 1$ two points with the same one-person set are the same point.	The one-person set.
8 $\frac{\exists}{\exists} = 1$	Make the set of a set, then read it back one level down.
9 fusion: $\text{Map}(f) \Rightarrow \frac{fR}{\exists} = f \frac{R}{\exists}$	Naturality of the unit, $\text{Map}(f) \Rightarrow f \frac{1}{\exists} = \frac{1}{\exists} E(f)$. renaming a point and then taking its one-person set is taking the set and renaming inside it.
10 $\text{Map}(f) \wedge 1\square = \exists\square \Rightarrow \frac{f}{\exists} = f \frac{1}{\exists}$	Rename first or take singletons first — the fusion row above at $R=1$.
11 $R\square = \exists\square \Rightarrow R/_s S = \frac{R}{\exists} (\frac{S}{\exists})^\circ$	x and y match when R sends x and S sends y to the same set.
12 $E(R) \triangleq \frac{\exists R}{\exists}$	$E(R) : EA \rightarrow EB$, image of a set of A
13 $\frac{R}{\exists} = \frac{1}{\exists} E(R)$	$\frac{1}{\exists} : x \mapsto \{x\}$
14 $\frac{S}{\exists} E(R) = \frac{SR}{\exists}$	absorption — the monad's composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{SR}{\exists}$, §sec-kleisli
15 $\supseteq \triangleq \exists/\exists$	$xs \supseteq ys \iff \forall a. ys \exists a \rightarrow xs \exists a$
16 $\sqsubseteq \triangleq \supseteq^\circ$	$xs \sqsubseteq ys \iff ys \supseteq xs$

§3.1.1.11. $i \dashv E$: power transpose through singleton

(3.1.1.11a)

§3.1.1.12. $E \setminus$ as a composite

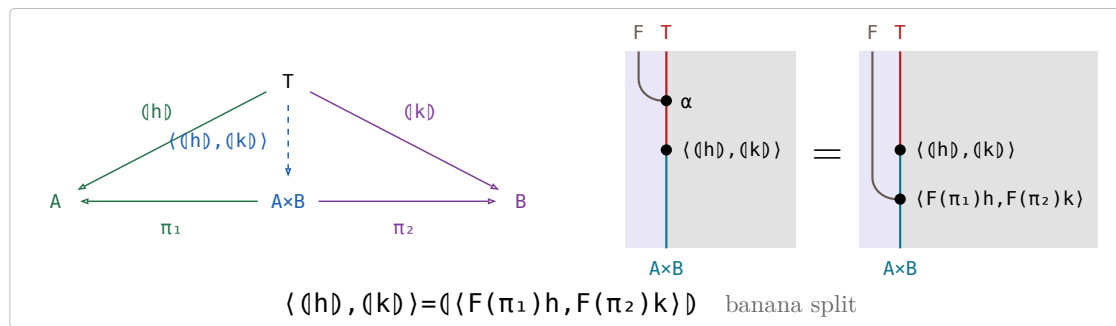
(3.1.1.12a)



(3.1.1.12b)

definition	type	note
$\text{listr } A ::= \text{nil} \mid \text{cons}(A, \text{listr } A)$	$\mathcal{A} \rightarrow \mathcal{A}$	The cons-lists over A , the datatype every row below folds.
$\text{sum} = ([\text{zero}, \text{plus}])$	$[A] \rightarrow A$	$\text{plus}(a, b) = a + b$.
$\text{length} = ([\text{zero}, \pi_2 \text{ succ}])$	$[A] \rightarrow \text{Nat}$	π_2 drops the head and keeps the count of the tail, succ adds one for the head.
$\text{average} \triangleq (\text{sum}, \text{length}) \text{div}$	$[\text{Nat}] \rightarrow \text{Nat}$	$\text{div}(m, n) = m/n$, with $\text{div}(0, 0) = 0$ so average is total. Traverses the list twice.
banana-split law $\langle \langle h \rangle, \langle k \rangle \rangle = \langle \langle F(\pi_1) h, F(\pi_2) k \rangle \rangle$	$T \rightarrow A \times B$	Any fork of folds is a single fold, hence one traversal — F the base functor.
what it reduces to $\alpha \langle \langle h \rangle, \langle k \rangle \rangle = F(\langle \langle h \rangle, \langle k \rangle \rangle) \langle F(\pi_1) h, F(\pi_2) k \rangle$	$FT \rightarrow A \times B$	All that $\triangleq$ reduce leaves to check: the fork satisfies the defining equation.
the instance $(\text{sum}, \text{length}) = ([\text{zeros}, \text{pluss}])$	$[\text{Nat}] \rightarrow \text{Nat} \times \text{Nat}$	$\text{pluss}(a, (b, n)) = (a + b, n + 1)$, so average runs in one pass.
$\text{preds } n = [n, n-1, \dots, 1]$	$\text{Nat} \rightarrow [\text{Nat}]$	apply the earlier special case to write preds as $\langle k \rangle \pi_1$.

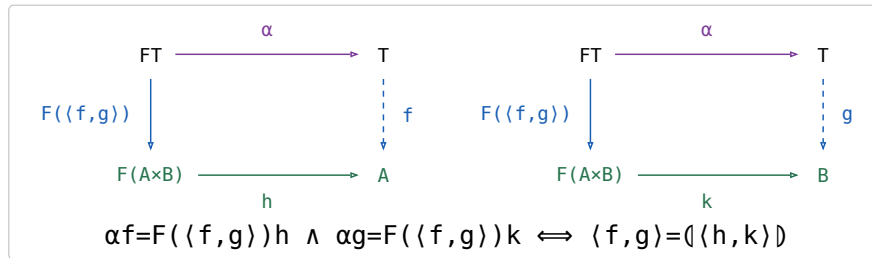
(3.1.1.12c)



§3.1.2. Fokkinga's mutual recursion theorem

- $F\text{-Alg}(\mathcal{A})$ — the algebras and their homomorphisms — has binary products whenever $\mathcal{A}$ does, and the forgetful $U : F\text{-Alg}(\mathcal{A}) \rightarrow \mathcal{A}$ creates them: the product of $h : FA \rightarrow A$ and $k : FB \rightarrow B$ is carried by $A \times B$, with structure $\langle F(\pi_1)h, F(\pi_2)k \rangle : F(A \times B) \rightarrow A \times B$.
- π_1 and π_2 are homomorphisms out of it, and $\langle p, q \rangle : X \rightarrow A \times B$ is a homomorphism **iff** p and q both are — substitute the structure and compare the two forks component by component.
- The banana-split law of (3.1.1.12b) IS that product's universal property read at the initial algebra: $\langle h \rangle$ and $\langle k \rangle$ are the unique homomorphisms to (A, h) and (B, k) , so their fork is the unique homomorphism into the product, which is $\langle \langle F(\pi_1)h, F(\pi_2)k \rangle \rangle$.
- Fokkinga's theorem is **strictly more general** and is not that product: in mutual recursion the two arrows leave $F(A \times B)$, not FA and FB , so each sees BOTH components. The product is the case where they factor as $F(\pi_1)h$ and $F(\pi_2)k$. What it says: **any algebra on a product carrier is folded by a fork**.

(3.1.2a)



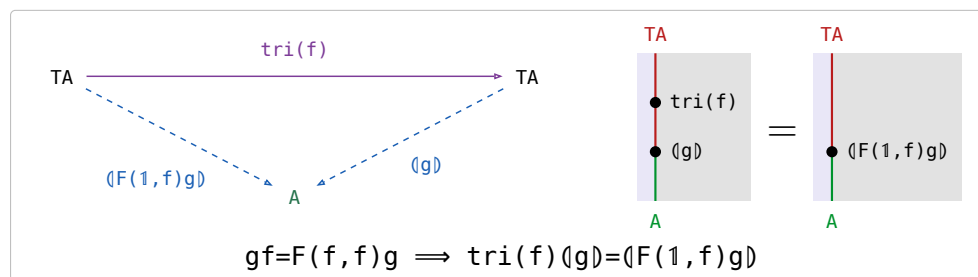
§3.2. Ruby triangles and Horner's rule

(3.2a)

stage	definition
informally	$\text{tri}(f)[a_0, a_1, \dots, a_i, \dots, a_n] = [a_0, f(a_1), \dots, f^i(a_i), \dots, f^n(a_n)]$
for cons-lists	$\text{tri}(f) = \langle [\text{nil}, (1 \times \text{listr}(f)) \text{ cons}] \rangle$
with the base functor named	$F(A, B) = 1 + A \times B$, $\alpha = [\text{nil}, \text{cons}]$, so $\text{tri}(f) = \langle F(1, \text{listr}(f)) \alpha \rangle$
abstractly	F a bifunctor with initial type (α, T) : $\text{tri}(f) = \langle F(1, T(f)) \alpha \rangle$

For the definition to make sense $f : A \rightarrow A$ is required, and then $\text{tri}(f) : TA \rightarrow TA$.

(3.2b)



Horner's rule is Horner's rule, generalised from cons-lists to any initial type. Fusion reduces it to $F(1, T(f)) \alpha (g) = F(1, (g)) F(1, f) g$.

§3.2.1. Depth of a tree

(3.2.1a)

the datatype	$\text{tree } A ::= \text{tip}(A) \mid \text{bin}(\text{tree } A, \text{tree } A)$, base functor $F(A,B)=A+B \times B$, initial type $([\text{tip}, \text{bin}], \text{tree})$
the fold	$\llbracket g, h \rrbracket$ is the unique f with $f(\text{tip}(a))=g(a)$ and $f(\text{bin}(x,y))=h(f(x), f(y))$
$\text{tree}(f)$	$\text{tree}(f) = \llbracket F(f, 1) \llbracket \text{tip}, \text{bin} \rrbracket \rrbracket$; pointwise $\text{tree}(f)(\text{tip}(a))=\text{tip}(f(a))$ and $\text{tree}(f)(\text{bin}(x,y))=\text{bin}(\text{tree}(f)(x), \text{tree}(f)(y))$
max	$\text{max} = \llbracket 1, \text{bmax} \rrbracket$, where $\text{bmax}(a,b)$ is the larger of a and b
depths	$\text{depths} = \text{tree}(\text{zero}) \text{ tri}(\text{succ})$ — replaces every tip by its depth in the tree, zero the constant function returning 0 and succ the successor
depth	$\text{depth} = \text{depths } \text{max}$

§3.3. The TeX problem

§3.4. Conditions and conditionals

§3.5. Concatenation and currying

If φ is natural, i.e. $G(h \times id) \cdot \varphi = \varphi \cdot (F(h) \times id)$, then $h \cdot G(f) \cdot \varphi = f \iff f = \text{apply} \cdot ((h \cdot G(\text{apply}) \cdot \varphi) \times id)$ Theorem 3.1
 The unique solution of a two-argument recursion $f(a, x) = h(G(f)(a, x))$ built from a natural φ is **apply** composed with a one-argument fold, which is how **cat** — not itself a fold — is derived from one.